

Course Syllabus

STI VISION: To be the leader in innovative and relevant education that nurtures individuals to become competent and responsible members of society		STI INSTITUTIONAL OUTCOMES:	
STI MISSION: We are an institution committed to provide knowledge through the development and delivery of superior learning systems.		Character (IO1): An STler is a person of character. An STler takes responsibility for his/her actions, treats people with respect, and lives with integrity.	
We strive to provide optimum value to all our stakeholders - our students, our faculty members, our employees, our partners, our shareholders, and our community.		Critical thinker (IO2): An STler is a critical thinker. An STler challenges and analyses all information through sound questioning and is unafraid to push for creative ideas.	
We will pursue this mission with utmost integrity, dedication, transparency, and creativity.		Communicator (IO3): An STler communicates to understand and be understood. An STler discerns the value of information read or heard and effectively expresses his/her own emotions when sharing information, may it be spoken or written.	
		Change-adept (IO4): An STler is change-adept. An STler can adjust, adapt, and reinvent continuously to changing circumstances. An STler believes in letting go of the old and embracing the new to achieve his/her fullest potential.	
SERIAL NUMBER: IT2511	SUBJECT TITLE: INFORMATION ASSURANCE AND SECURITY (DATA PRIVACY)		CREDIT: 2 lec, 1 lab (2 hrs. lec and 3 hrs. lab per week)
COURSE DESCRIPTION:	This course covers the principles of data privacy. Students will gain insight into vulnerability analysis, counter-attack strategies, and other advanced information assurance and security concepts.		
PREREQUISITE:	Information Management		
COURSE OUTCOMES:	After successful completion of this course, the student should be able to: CO1. Examine various reconnaissance techniques by collecting and processing information from networks, wireless systems, IoT traffic, and target hosts; (PI2 – ITPO04) CO2. Assess and validate the security posture of mobile, OT, and organizational environments using appropriate security tools, threat models, and professional standards; and (PI3 – ITPO05) CO3. Design controlled exploitation procedures, including payload generation, cloud misconfiguration abuse, and methodology-guided attack simulations, to demonstrate offensive cybersecurity capability. (PI4 – ITPO07)		
MANDATED BOOKS:	None		
REFERENCE:	1. EC-Council. (2021). <i>Ethical hacking essentials (1st ed.)</i> . EC-Council.		
COURSE REQUIREMENTS:	• Class Participation • Seatwork/Exercises • Quizzes and Summative Tests • Group Work		
GRADING SYSTEM:	The following percentage distribution shall be followed: Prelims 20% Midterms 20%		

Pre-finals	20%
<u>Finals</u>	<u>40%</u>
	100%

The following are the periodical grade components for this course:

Quizzes	20%
Performance Task	30%
<u>Major Examination</u>	<u>50%</u>
	100%

Course Outline

Learning Outcomes (LO)		Week	LEC Hours	LAB Hours	TOPICS	Teaching and Learning Activities	Assessment Tasks
1	Conduct a Wi-Fi packet analysis using wireless security tools (CO1)	1-2	4	6	Wireless Attacks and Countermeasures Wireless Encryption Algorithms Different Wireless Attack Tools and Countermeasures Bluetooth Attacks Different Wireless Security Tools	Lecture 01 Seatwork 1	01 Quiz 1 01 Laboratory Exercise 1
2	Create binary payloads to hack an Android device (CO3)	3-4	4	6	Mobile Attacks and Countermeasures Anatomy of a Mobile Attack Mobile Platform Attack Vectors and Vulnerabilities Mobile Security Guidelines and Security Tools	Lecture	02 Quiz 1
3	Assess Android devices using various Android security tools (CO2)						02 Laboratory Exercise 1 02 Performance Task 1
		5			PRELIMINARY EXAMINATION		Pen-and-Paper Test
4	Perform footprinting using various footprinting techniques (CO1)	6-7	4	6	IoT Attacks and Countermeasures IoT Concepts IoT Attacks IoT Attack Countermeasures	Lecture	03 Laboratory Exercise 1
5	Analyze IoT traffic using a network packet analyzer (CO1)						03 Quiz 1 03 Laboratory Exercise 2
6	Evaluate OT concepts based on real-life threats (CO2)	8-9	4	6	OT Attacks and Countermeasures OT Concept OT Threats and Attacks OT Attack Countermeasures	Lecture 04 Group Activity 1	04 Quiz 1 04 Performance Task 1
		10			MIDTERM EXAMINATION		Pen-and-Paper Test

7	Use enumeration tools to collect information about the target system (CO1)	11-13	6	9	Cloud Computing Threats and Countermeasures Cloud Computing Concepts Container Technology Cloud Computing Threats, Tools, and Countermeasures	Lecture 05 Seatwork 1	05 Quiz 1 05 Laboratory Exercise 1 05 Quiz 2 05 Performance Task 1
8	Exploit Open S3 Buckets to compromise data privacy (CO3)						
		14			PRE-FINAL EXAMINATION		Pen-and-Paper Test
9	Apply the penetration testing methodologies to cybersecurity scenarios (CO3)	15-17	6	9	Penetration Testing Fundamentals Types of Penetration Testing Phases of Penetration Testing Penetration Testing Methodologies Guidelines and Recommendations for Penetration Testing <i>EHE Certification Exam</i>	Lecture 06 Group Activity 1	06 Quiz 1 06 Laboratory Exercise 1 06 Quiz 2 06 Performance Task 1
10	Assess the different guidelines and recommendations for a penetration tester (CO2)						
		18			FINAL EXAMINATION		Pen-and-Paper Test

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